Chapter 17 Notes

Section 1

Notes

Here is an explanation for the remarks given on page 224. Note that I am assuming M contains all ordinals (see the section for Remark 3). Definition 1.1 to even work in the first place. If a function f has domain α, and is such that, for all $\beta < \alpha$, $f(\beta) = R_\beta^{\mathcal{G}}$, call f an $R_\alpha^{\mathcal{G}}$ function. Let R-alpha func(f) stand for the following formula:

$$\text{Func}(f) \wedge \text{Dom}(f) = \alpha \wedge f(0) = 0 \wedge$$

$$(\forall \beta < \alpha)(\beta^+ < \alpha \supset (\forall x)(x \in f(\beta^+) \equiv x \subseteq \mathcal{G} \times f(\beta)))$$

$$\wedge (\forall \beta < \alpha)(\text{beta is a successor ordinal} \supset (\forall x)(x \in f(\beta)$$

$$\equiv (\exists \delta < \beta)(x \in f(\delta))))$$

Note that R-alpha func(f) just means f is the $R_\alpha^{\mathcal{G}}$ function.

Lemma 0.1. For any α, if, $R_\beta^{\mathcal{G}} \in M$ for all $\beta < \alpha$, then both $R_\alpha^{\mathcal{G}} \in M$ and the $R_\alpha^{\mathcal{G}}$ function are in M.

Proof. By transfinite induction.

The least ordinal case is obvious (because $\emptyset \in M$).

Suppose it holds for all ordinals less than some ordinal α. So suppose that $R_\beta^{\mathcal{G}} \in M$ for all $\beta < \alpha$. Then it follows that the $R_\beta^{\mathcal{G}}$ function is in M for all $\beta < \alpha$. We first show that $R_\alpha^{\mathcal{G}} \in M$. Well, if α is a successor ordinal of some β, then it is defined by the formula

$$(\forall y)(y \in x \equiv y \subseteq \mathcal{G} \times R_\beta^{\mathcal{G}})$$

which works, as $\mathcal{G} \times R_\beta^{\mathcal{G}} \in M$ for the following reason: both $\mathcal{G}$ and $R_\beta^{\mathcal{G}}$ are in M, meaning $\{\mathcal{G}, R_\beta^{\mathcal{G}}\} \in M$ (because M is a first order universe). Thus $\cup\{\mathcal{G}, R_\beta^{\mathcal{G}}\} \in M$, meaning $\mathcal{P}(\cup\{\mathcal{G}, R_\beta^{\mathcal{G}}\}) \cap M \in M$, and thus (1) $\mathcal{P}(\mathcal{P}(\cup\{\mathcal{G}, R_\beta^{\mathcal{G}}\}) \cap M) \cap M \in M$. Now, it is obvious that the formula in M in (1) is such that $\langle x, y\rangle$ is in it for each $x \in \mathcal{G}$ and each $y \in R_\beta^{\mathcal{G}}$. Then it is easy to see that $\mathcal{G} \times R_\beta^{\mathcal{G}}$ is first-order definable over M, and, because this would mean it is a first-order definable susbset of the set in (1), $\mathcal{G} \times R_\beta^{\mathcal{G}} \in M$ (because M is first-order swelled).

The case where α is a limit ordinal is mostly similar. In particular, because we are assuming the $R_\beta^{\mathcal{G}}$ function is in M for each $\beta < \alpha$, the following defines $R_\alpha^{\mathcal{G}}$:

$$(\exists \beta < \alpha)(\exists f)(\text{R-beta func}(f) \wedge (\exists \delta < \beta)(x \in f(\delta)))$$

Thus we know that $R_\alpha^{\mathcal{G}} \in M$.

Now we show that the $R_\alpha^{\mathcal{G}}$ function is definable over M. Well, if α is a successor ordinal of some β, then the $R_\alpha^{\mathcal{G}}$ function is definable over M by the formula

$$(x = \beta \wedge y = R_\beta^{\mathcal{G}}) \vee$$

$$(\exists \delta < \alpha)(\exists f)(\text{R-delta func}(f) \wedge x \in \text{Dom}(f) \wedge y = f(x))$$

and if α is a limit ordinal, then it is defined over M by the following formula

$$(\exists \delta < \alpha)(\exists f)(\text{R-delta func}(f) \wedge x \in \text{Dom}(f) \wedge y = f(x))$$

Thus, in any case, we know that the $R_\alpha^{\mathcal{G}}$ function is definable over M. Because $\alpha \in M$, this means that the range of the $R_\alpha^{\mathcal{G}}$ function, the set S of all $R_\beta^{\mathcal{G}}$ for $\beta < \alpha$, is in M. Hence $\{\alpha, S\} \in M$, $\cup\{\alpha, S\} \in M$, $\mathcal{P}(\cup\{\alpha, S\}) \in M$, and $\mathcal{P}(\mathcal{P}(\cup\{\alpha, S\})) \in M$. Now, for each $\beta < \alpha$ and $x \in S$, it of course follows that $\langle \beta, x\rangle \in \mathcal{P}(\mathcal{P}(\cup\{\alpha, S\}))$. That is, the $R_\alpha^{\mathcal{G}}$ function is a first-order definable subset of $\mathcal{P}(\mathcal{P}(\cup\{\alpha, S\}))$, which is in M, meaning the function itself is in M. □

Remark 1

The fact that each $R_\alpha^{\mathcal{G}} \in M$ is immediate from the above lemma (using an obvious transfinite induction argument). Then $\mathcal{D}^{\mathcal{G}}$ being a subclass of M follows because M is transitive.

Remark 2

We show that $R_\alpha^{\mathcal{G}} \subseteq R_{\alpha+1}^{\mathcal{G}}$. In fact, we directly prove the stronger result, that $\beta < \alpha$ implies $R_\beta^{\mathcal{G}} \subseteq R_\alpha^{\mathcal{G}}$.

Proof. By transfinite induction.

The least ordinal case is vacuous.

Suppose the result holds for some α. Now, if $x \in R_\alpha^{\mathcal{G}}$, there is some $\beta < \alpha$ such that all members of x are of the form $\langle y, z\rangle$ where $y \in \mathcal{G}$ and $z \in R_\beta^{\mathcal{G}}$. By the induction assumption, this means that $z \in R_\alpha^{\mathcal{G}}$. Then $\langle y, z\rangle$ would also be in $R_\alpha^{\mathcal{G}}$ for all such y and z, and then so would x (as we are assuming x is in M).

The limit ordinal case is obvious. □

Remark 3

We first prove the following lemma.

Lemma 0.2. For each ordinal α, if α is the least ordinal or a limit ordinal, $R_\alpha^{\mathcal{G}} \in R_{\alpha^+}^{\mathcal{G}}$ (note, however, that $R_{\alpha^+}^{\mathcal{G}} \notin R_{\alpha^{++}}^{\mathcal{G}}$ in general for successor ordinals α^+).

Proof. The least ordinal case is obvious, as $R_0^{\mathcal{G}} \in R_1^{\mathcal{G}}$ because $R_0^{\mathcal{G}} = \emptyset$ and $R_1^{\mathcal{G}} = \{\emptyset\}$.

Consider some limit ordinal λ. Hence $R_\lambda^{\mathcal{G}} = \cup_{\alpha<\lambda} R_\alpha^{\mathcal{G}}$. We already know that $R_\lambda^{\mathcal{G}}$ is in M (by (1) of the remarks). Also, it can easily be seen that $R_\lambda^{\mathcal{G}} \subseteq \mathcal{G} \times R_\lambda^{\mathcal{G}}$. Thus, by definition of $R_{\lambda^+}^{\mathcal{G}}$, it follows that $R_\lambda^{\mathcal{G}} \in R_{\lambda^+}^{\mathcal{G}}$. □

We now prove that $R_{\alpha+1}^{\mathcal{G}} \neq R_\alpha^{\mathcal{G}}$.

Proof. By transfinite induction. In particular, we show that, for every α, $R_\beta^{\mathcal{G}} \neq R_\alpha^{\mathcal{G}}$ for every $\beta < \alpha$.

Suppose the result holds for all ordinals less than some ordinal δ.

Least ordinal case: The case when $\delta = \emptyset$ is vacuous.

Successor ordinal case: Suppose δ is the successor of some ordinal α. There are two cases: either α itself is a successor ordinal, or it is a limit ordinal (of course, α could be $\emptyset$, but that case is trivial). We must show the result holds for α^+ (that is, δ). Note that it suffices to show that $R_{\alpha+1}^{\mathcal{G}} \neq R_\alpha^{\mathcal{G}}$, for, if we show that, then if $R_{\alpha+1}^{\mathcal{G}} = R_\beta^{\mathcal{G}}$ for some $\beta < \alpha$, then it would follow that $R_\beta^{\mathcal{G}} = R_\alpha^{\mathcal{G}}$ by part (2) of the remarks, contrary to the induction assumption.

On the one hand, suppose that α itself is a successor ordinal (say of some β). Thus, by the assumption, we know that there must be some $y \in R^{\mathcal{G}}_{\alpha}$ such that $y \notin R^{\mathcal{G}}_{\beta}$. Consider any $x \in \mathcal{G}$. Then $\{\langle x, y\rangle\} \in R^{\mathcal{G}}_{\alpha+1}$ (because M is a first-order universe). If $R^{\mathcal{G}}_{\alpha+1} = R^{\mathcal{G}}_{\alpha}$, then it would follow that $z = \{\langle x, y\rangle\}$ for some $z \in R^{\mathcal{G}}_{\alpha}$. What this would mean is that $\langle x, y\rangle \in \mathcal{G} \times R^{\mathcal{G}}_{\beta}$. But that would mean $y \in R^{\mathcal{G}}_{\beta}$, a contradiction. Thus it must be that $R^{\mathcal{G}}_{\alpha+1} \neq R^{\mathcal{G}}_{\alpha}$.

On the other hand, suppose that α is a limit ordinal. By Lemma 0.2, it follows that $R^{\mathcal{G}}_{\alpha} \in R^{\mathcal{G}}_{\alpha^+}$. If $R^{\mathcal{G}}_{\alpha^+} = R^{\mathcal{G}}_{\alpha}$, then $R^{\mathcal{G}}_{\alpha} \in R^{\mathcal{G}}_{\alpha}$, which contradicts the assumption that M, and, by extension, $R^{\mathcal{G}}_{\alpha}$, is well founded.

Limit ordinal case: This is easy. □

Now we show that $\mathcal{D}^{\mathcal{G}}$ is a proper class in M (that is, $\mathcal{D}^{\mathcal{G}} \notin M$). Suppose it were in M. Then, because $R^{\mathcal{G}}_0, R^{\mathcal{G}}_1, ..., R^{\mathcal{G}}_{\alpha}, ...$ is an ordinal hierarchy, it follows by Theorem 2.1 of Chapter 7 that there is some ordinal α such that $\mathcal{D}^{\mathcal{G}} \subseteq R^{\mathcal{G}}_{\alpha}$. By (2), $\mathcal{D}^{\mathcal{G}} \subseteq R^{\mathcal{G}}_{\alpha+1}$. Also $R^{\mathcal{G}}_{\alpha+1} \subseteq \mathcal{D}^{\mathcal{G}}$, meaning $R^{\mathcal{G}}_{\alpha+1} = \mathcal{D}^{\mathcal{G}}$. But this contradicts the fact that $R^{\mathcal{G}}_{\alpha+1} \neq R^{\mathcal{G}}_{\alpha}$. Thus it cannot be that $\mathcal{D}^{\mathcal{G}} \in M$.

The above paragraph explains why I am assuming contains all ordinals. For if it didn't, then we wouldn't even know the sequence from Definition 1.1 was defined for all ordinals, and thus we couldn't use Theorem 2.1 of Chapter 7. However, it may indeed be that we could prove the result without this theorem - I was, however, unable to see this.

Remark 4

Suppose $f \subseteq \mathcal{G} \times \mathcal{D}^{\mathcal{G}}$ and f is an *M-set*. Now, for each $x \in f$, x is of the form $\langle y, z\rangle$ for some $y \in \mathcal{G}$ and some $z \in R^{\mathcal{G}}_{\alpha}$ for some α. Let $o(f)$ be the class of all α such that $\langle x, y\rangle \in f$ for at least one $y \in R^{\mathcal{G}}_{\alpha}$. We show that there must be some β higher than all the $\alpha \in o(f)$. For suppose there weren't. Then it would follow that $o(f)$ could be put into a 1-1 correspondence with On. But $o(f)$ is a set (because f is a set), which would mean that On is a set (by the axiom of substitution), which is impossible. Thus it must be that there is some β higher than all the $\alpha \in o(f)$. Consider the least such β. Because $f \in M$, it is easy to see that $f \in R^{\mathcal{G}}_{\beta+1}$, and thus $f \in \mathcal{D}^{\mathcal{G}}$.

Remark 5

I could not see how to show the isomorphism, but the rest of the remark seems to be just Exercise 1.2, which I will give here.

Here is how we could get this recursion to work in the first place. We define a function m_{α} for each α whose domain is $R^{\mathcal{G}}_{\alpha}$ and which satisfies the conditions of the function. We use transfinite recursion, where $m_{\alpha+1} = g(m_{\alpha})$, where $g(m_{\alpha})$ is the set containing all members of m_{α}, as well as all sets of the form $\langle x, \{m_{\alpha}(g) | \langle p, g\rangle \in x\}\rangle$ for $x \in R^{\mathcal{G}}_{\alpha+1} - R^{\mathcal{G}}_{\alpha}$.

Suppose $\mathcal{G} = \{p\}$, and a mapping m is defined by

$$m(f) = \{m(g) | \langle p, g\rangle \in f\}$$

First, we show that the mapping is 1-1. Note for this that an obvious transfinite induction argument shows that if $\beta < \alpha$, $m(g)$ stays constant between m applied to $R^{\mathcal{G}}_{\beta}$ and $R^{\mathcal{G}}_{\alpha}$ (that is, between m_{β} and m_{α}), for any $g \in R^{\mathcal{G}}_{\beta}$.

Proof. By transfinite induction.

The case for $R^{\mathcal{G}}_0$ obviously holds.

Suppose it holds for all ordinals less than some α. Now consider any f and h in $R^{\mathcal{G}}_{\alpha}$. Suppose $m(f) = m(h)$. Then

$$m(f) = \{m(g) | \langle p, g\rangle \in f\} = m(h) = \{m(l) | \langle p, l\rangle \in f\}$$

Then it easily follows by the induction assumption that $f = h$, as the values of $m(g)$ for all $g \in R^{\mathcal{G}}_{\beta}$ for some $\beta < \alpha$ stay constant. □

Next we show that, for all $f \in R_\alpha^{\mathcal{G}}$, $m(f) \in R_\alpha$.

Proof. By transfinite induction.

The case for $R_0^{\mathcal{G}}$ vacuously holds.

Suppose it holds for some α. Consider any $x \in R_{\alpha+1}^{\mathcal{G}}$; then $m(f) = \{m(g) | \langle p, g\rangle \in f\}$, and, by assumption, each $m(g)$ would be in R_α. Thus $m(f)$ is a subset of R_α, and, consequently, it is in $R_{\alpha+1}$ (because $R_{\alpha+1} = \mathcal{P}(R_\alpha)$.

The limit ordinal case is obvious. □

The last part of the exercise, showing that if $x \in R_\alpha$, then there is some $y \in R_\alpha^{\mathcal{G}}$ such that $m(y) = x$, was one the one I was unable to show.

Clarification

Definition 1.4 in the book needs clarification. In particular, the $[\![]\!]$ translation is being used, yet this has not yet been defined for the $\approx_\alpha$ relation. The definition should be: $[\![f \approx_{\alpha+1} g]\!] = f \approx_{\alpha+1} g$, $[\![f \approx_\lambda g]\!] = \Box\Diamond f \approx_\lambda$ for λ a limit ordinal, and $[\![x \epsilon y]\!] = \Box\Diamond x \epsilon y$.

The errata listed at `https://id144254.securedata.net/melvinfitting/errata/book_errors/RevisedSetBookErrors/RevisedSetCorrections.pdf` are important to note for this section.

It should be noted why $(f \approx_\lambda g) \supset [\![f \approx_\lambda g]\!]$ for limit ordinals λ. We prove this using transfinite induction. Suppose it holds for all ordinals less than λ. By the definition of $\approx_\lambda$, $(f \approx_\alpha g) \supset (f \approx_\lambda g)$ is valid in the model for all $\alpha < \lambda$. Thus $\Box\Diamond(f \approx_\alpha g) \supset \Box\Diamond(f \approx_\lambda g)$ is also valid in the model, that is, $[\![f \approx_\alpha g]\!] \supset [\![f \approx_\lambda]\!]$ is valid in the model. Now if $p \vDash (f \approx_\lambda g)$, then $p \vDash (f \approx_\alpha g)$ for some $\alpha < \lambda$, and by the induction assumption, this means $p \vDash [\![f \approx_\alpha g]\!]$, hence $p \vDash [\![f \approx_\lambda g]\!]$.

Proposition 1.5

One such error listed in the errata is with regards to the limit ordinal case in this proposition. The definition of $\approx_\lambda$ is not that $p \vDash [\![f \approx_\lambda]\!]$ implies $p \vDash [\![f \approx_\alpha]\!]$ for some $\alpha < \lambda$, but rather that $p \vDash f \approx_\lambda$ implies $p \vDash f \approx_\alpha$ for some $\alpha < \lambda$. Thus the inference made in the proof is invalid.

Instead, we show that $p \vDash f \approx_\lambda g$ implies $p \vDash [\![f \approx_\beta g]\!]$. Well, if $p \vDash f \approx_\lambda g$, then $p \vDash f \approx_\alpha g$ for some $\alpha < \lambda$. Then it easily follows that $p \vDash [\![f \approx_\alpha g]\!]$, and thus, by the induction assumption, $p \vDash [\![f \approx_\beta g]\!]$.

Thus $p \vDash (f \approx_\lambda g) \supset [\![f \approx_\beta g]\!]$ is valid in the model. Hence, using standard $S4$ manipulation, $p \vDash \Box\Diamond(f \approx_\lambda g) \supset \Box\Diamond[\![f \approx_\beta g]\!]$, that is, $p \vDash [\![f \approx_\lambda g]\!] \supset [\![f \approx_\beta g]\!]$, is valid in the model.

Exercise 1.3

We show that for $f, g \in R_\alpha^{\mathcal{G}}$, if $p \vDash (f \approx g)$, then $p \vDash (f \approx_\alpha g)$.

Proof. By transfinite induction. In particular, we make the induction assumption that, for all possible worlds p (we are not just considering one particular one), if $p \vDash (f \approx g)$ for $f, g \in R_\alpha^{\mathcal{G}}$, then $p \vDash (f \approx_\alpha g)$.

This vacuously holds for the case of 0, and the limit ordinal case is obvious.

The successor ordinal case is where the real work is required. Suppose the result holds for all ordinals less than some successor ordinal α^+. Suppose $f, g \in R_{\alpha^+}^{\mathcal{G}}$, and $p \vDash (f \approx g)$. Now, this means $p \vDash (f \approx_\beta g)$ for some β. This means

$p \vDash (f \approx_{\delta^+} g)$ for some successor ordinal $\delta^+ \leq \beta$. Thus we have

$$p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\delta x)] \wedge$$

$$(\forall x)[(x \epsilon g) \supset (\exists y)(y \epsilon f \wedge y \approx_\delta x)]]\!]$$

meaning we have

$$p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\delta x)]]\!]$$

which means we have, for each $d \in \mathcal{D}^{\mathcal{G}}$,

$$p \vDash [\![(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\delta x)]\!]$$

meaning we have

$$(1)\ p \vDash \Box\Diamond([\![x \epsilon f]\!] \supset [\![(\exists y)(y \epsilon g \wedge y \approx_\delta d))]\!]$$

Because we want to show $p \vDash (f \approx_{\alpha^+} g)$, what we aim to show is

$$p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha x)] \wedge$$

$$(\forall x)[(x \epsilon g) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha x)]]\!]$$

It suffices to show both $p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha x)]]\!]$ and $p \vDash [\![(\forall x)[(x \epsilon g) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha x)]]\!]$. We give the proof of (2) $p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha x)]]\!]$ - the other case is similar. Now, (2) is the same as saying $p \vDash (\forall x)[\![(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha x)]\!]$. It suffices to show that, for all $d \in \mathcal{D}^{\mathcal{G}}$, $p \vDash [\![(d \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha d)]\!]$. So consider any $d \in \mathcal{D}^{\mathcal{G}}$. We must show that $p \vDash [\![(d \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha d)]\!]$, which is equivalent to $p \vDash \Box\Diamond([\![d \epsilon f]\!] \supset [\![(\exists y)(y \epsilon g \wedge y \approx_\alpha d)]\!])$, which is equivalent to $(2')\ p \vDash \Box\Diamond([\![d \epsilon f]\!] \supset \Box\Diamond(\exists y)[\![y \epsilon g \wedge y \approx_\alpha d]\!])$. To show this, consider any p' accessible from p. By (1), there is some p'' accessible from p' such that $p'' \vDash [\![x \epsilon f]\!] \supset [\![(\exists y)(y \epsilon g \wedge y \approx_\delta d)]\!]$, which is equivalent to (3) $p'' \vDash [\![d \epsilon f]\!] \supset \Box\Diamond[\![(\exists y)(y \epsilon g \wedge y \approx_\delta d)]\!]$. We show that (4) $p'' \vDash [\![d \epsilon f]\!] \supset \Box\Diamond(\exists y)[\![y \epsilon g \wedge y \approx_\alpha d]\!]$ (which would imply $(2')$). So suppose that $p'' \vDash [\![d \epsilon f]\!]$. By (3), this means that (5) $p'' \vDash \Box\Diamond[\![(\exists y)(y \epsilon g \wedge y \approx_\delta d)]\!]$. To show that (4) holds, consider any q accessible from p''. By (5), there is some q' accessible from q such that $q' \vDash [\![(\exists y)(y \epsilon g \wedge y \approx_\delta d)]\!]$. Then there is some q'' accessible from q' such that there is some $k \in \mathcal{D}^{\mathcal{G}}$ with $q'' \vDash [\![k \epsilon g \wedge k \approx_\delta d]\!]$, which implies both $q'' \vDash [\![k \epsilon g]\!]$ and $q'' \vDash [\![k \approx_\delta d]\!]$. Note that $d, k \in R^{\mathcal{G}}_\alpha$. Because $q'' \vDash [\![k \approx_\delta d]\!]$, it follows that $q'' \vDash \Box\Diamond(k \approx_\delta d)$. Thus there is some r' accessible from some r accessible from q'' such that $r' \vDash (k \approx_\delta d)$. Now, this implies $r' \vDash (k \approx d)$. By the induction assumption, this tells us that $r' \vDash (k \approx_\alpha d)$, which implies $r' \vDash [\![k \approx_\alpha]\!]$. Also, because r is accessibe from q'' (by the transitivity of the accessibility relation), and $q'' \vDash [\![k \epsilon g]\!]$, $r' \vDash [\![k \epsilon g]\!]$. Now, because r' is accessible from q, the fact that we have shown both $r' \vDash [\![k \approx_\alpha]\!]$ and $r' \vDash [\![k \epsilon g]\!]$ proves that $p'' \vDash \Box\Diamond(\exists y)[\![(y \epsilon g) \wedge (y \approx_\alpha d)$, which completes the proof in this case. □

Defintion 1.7

It was not entirely clear to me how $[\![f \in g]\!]$ was meant to be defined, but it seems to me like it would just consist of adding $\Box\Diamond$ to the beginning of Definition 1.7.

Section 2

Lemma 2.1

It is claimed that $(f \approx_\alpha g) \equiv [\![f \approx_\alpha g]\!]$, yet this is incorrect. In particular, if $[\![(f \approx_\alpha g)]\!]$ for α a limit ordinal, it does not necessarily follow that $(f \approx_\alpha g)$.

The proof of this lemma leaves out the limit ordinal case. Here is an explanation for that case. Suppose the result holds for all $\alpha < \lambda$ for some limit ordinal λ. Now suppose that (1) $p \vDash [\![f \approx_\lambda g]\!]$ and (2) $p \vDash [\![g \approx_\lambda h]\!]$, which imply both $p \vDash \Box\Diamond(f \approx_\lambda g)$ and $p \vDash \Box\Diamond(g \approx_\lambda h)$. To prove $p \vDash [\![f \approx_\lambda h]\!]$, it suffices to show that $p \vDash \Box\Diamond(f \approx_\lambda h)$. So consider any p' accessible from p. By (1), this means there is some p'' accessible from p' such that $p'' \vDash f \approx_\lambda g$, meaning $p'' \vDash f \approx_\alpha g$ for some $\alpha < \lambda$, and thus (3) $p'' \vDash [\![f \approx_\alpha g]\!]$. Because p'' is itself accessible from p, it follows by (2) that there is some p''' accessible from p'' such that $p''' \vDash g \approx_\lambda h$, which implies $p''' \vDash g \approx_\beta h$ for some $\beta < \lambda$, which implies $p''' \vDash [\![g \approx_\beta h$. Now, by (3), and the fact that p''' is accessible from p'', $p''' \vDash [\![f \approx_\alpha g]\!]$. Pick the greater of α and β (say it is α). Then, by cumulativity, $p''' \vDash [\![g \approx_\alpha h]\!]$. Hence, by the induction assumption, it follows that $p''' \vDash [\![f \approx_\alpha h]\!]$. What this means is that for some q' accessible from some q accessible from p''', $q' \vDash f \approx_\alpha h$, which implies $q' \vDash f \approx_\lambda h$ (because $\alpha < \lambda$). But q' is accessible from p'. Thus we have shown, for every world p' accessible from p, there is some world q' such that $q' \vDash f \approx_l ambdah$ holds, and this proves $p \vDash \Box\Diamond(f \approx_\lambda h)$, that is, $p \vDash [\![f \approx_\lambda h]\!]$.

Theorem 2.2

The proof of this theorem is incorrect. It is claimed that $p \vDash [\![f \approx g]\!]$ implies $p \vDash [\![f \approx_\alpha g]\!]$ for some α, but this may not be the case. Here is a corrected proof (it is in fact reminiscent of the proof of the limit ordinal case of Lemma 2.1).

Suppose $p \vDash [\![f \approx g]\!]$, that is, (3) $p \vDash \Box\Diamond(f \approx g)$. Also suppose $p \vDash [\![g \approx h]\!]$, that is, (2) $p \vDash \Box\Diamond(g \approx h)$. To show $p \vDash [\![f \approx h]\!]$, we show $p \vDash \Box\Diamond(f \approx h)$. So consider any p' accessible from p. Then, by (2), there is some p'' accessible from p' such that $p'' \vDash f \approx g$, meaning $p'' \vDash f \approx_\alpha g$ for some α, hence (3) $p'' \vDash [\![f \approx_\alpha g]\!]$. Because p'' is itself accessible from p, (2) tells us that $p''' \vDash g \approx h$ for some p''' accessible from p'', which means $p''' \vDash g \approx_\beta h$ for some β, which implies $p''' \vDash [\![g \approx_\beta h]\!]$. Because p''' is accessible form p'', we have $p''' \vDash [\![f \approx_\alpha g]\!]$ by (3). Without loss of generality, let α be the maximum of α and β. Then, by cumulativity, $p''' \vDash [\![g \approx_\alpha h]\!]$. Hence, by Lemma 2.1, it follows that $p''' \vDash [\![f \approx_\alpha h]\!]$. This means $q' \vDash f \approx_h$ for some q' accessible from some q accessible from p''', which implies $q' \vDash f \approx h$. But q' is then accessible from p'. Thus we have shown that, for every p' accessible from p, there is some q' accessible from p' such that $q' \vDash f \approx h$, which proves that $p \vDash \Box\Diamond(f \approx h)$, that is, $p \vDash [\![f \approx h]\!]$.

Exercise 2.1

We first prove that if $f \in R^{\mathcal{G}}_\alpha$, then $p \vDash (f \approx_\alpha f)$.

Proof. By transfinite induction. The least ordinal case is vacuous, and the limit ordinal case is obvious.

We now consider the successor ordinal case. So suppose the result holds for some α, and suppose $f \in R^{\mathcal{G}}_{\alpha^+}$. We must show $p \vDash (f \approx_{\alpha^+} f)$, that is,

$$p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha x)] \wedge$$
$$(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha x)]]\!]$$

which is the same as just showing $p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha x)]]\!]$. So consider any $d \in \mathcal{D}^{\mathcal{G}}$. We must show that $p \vDash [\![(d \epsilon f) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha d)]\!]$, that is, $p \vDash \Box\Diamond([\![d \epsilon f]\!] \supset [\![(\exists y)(y \epsilon f \wedge y \approx_\alpha d)]\!])$. So consider any p' with $p \mathcal{R} p'$. Then consider any p'' with $p' \mathcal{R} p''$. We show that $p'' \vDash ([\![d \epsilon f]\!] \supset [\![(\exists y)(y \epsilon f \wedge y \approx_\alpha d)]\!])$. Suppose $p'' \vDash [\![d \epsilon f]\!]$. This means that, for some q' accessible from some q accessible from p'', $q' \vDash d \epsilon f]\!]$, which implies that $d \in R^{\mathcal{G}}_\alpha$. Now, to show that $p'' \vDash [\![(\exists y)(y \epsilon f \wedge y \approx_\alpha d)]\!]$, we show that $p'' \vDash [\![(d \epsilon f \wedge d \approx_\alpha d)]\!]$, that is, that both $p'' \vDash [\![d \epsilon f]\!]$ and $p'' \vDash [\![d \approx_\alpha d]\!]$. We already know that $p'' \vDash [\![d \epsilon f]\!]$. As for $p'' \vDash [\![d \approx_\alpha d]\!]$, this is obvious now that we know $d \in R^{\mathcal{G}}_\alpha$ (because, by the induction assumption, this means that $r \vDash d \approx_\alpha d$ for all possible worlds r). Thus we have $p'' \vDash [\![(d \epsilon f \wedge d \approx_\alpha d)]\!]$, hence $p'' \vDash [\![(\exists y)(y \epsilon f \wedge y \approx_\alpha d)$, which is what we wanted to show. □

Thus, we know that $f \in R^{\mathcal{G}}_\alpha$ implies $p \vDash (f \approx_\alpha f)$, hence $p \vDash (f \approx f)$, hence $p \vDash [\![(f \approx f)]\!]$.

Lemma 2.6

There are two problems in this proof. First, $[\![f \approx g]\!]$ being true at p does not mean $[\![f \approx_{\alpha+1} g]\!]$ is true at p for some α, but rather, $[\![f \approx_{\alpha+1} g]\!]$ is true at some q accessible from p. Similarly, $p'' \vDash [\![b \approx_\alpha a]\!]$ does not mean $p'' \vDash [\![b \approx a]\!]$, but rather $r \vDash [\![b \approx a]\!]$ for some r accessible from p''. These two things are not major issues in the proof, and can be easily corrected.

Exercise 2.2

We prove that $[\![(\forall x)(\forall y)[(x \epsilon y) \supset (\phi(x) \equiv \phi(y))]]\!]$ is $S4$ valid in $\mathcal{M}$. It suffices to show that, for all $f, g \in \mathcal{D}^{\mathcal{G}}$, and any possible world p, $p \vDash [\![(d \epsilon f) \supset (\phi(d) \equiv \phi(f))$, that is, $p \vDash \Box\Diamond([\![d \epsilon f]\!] \supset [\![(\phi(d) \equiv \phi(f))]\!])$. So consider anf p' with $p \mathcal{R} p'$ and anf p'' with $p' \mathcal{R} p''$. We show $p'' \vDash ([\![d \epsilon f]\!] \supset [\![(\phi(d) \equiv \phi(f))]\!])$. Suppose $p'' \vDash [\![d \epsilon f]\!]$. For the sake of contradiction, suppose that $p'' \nvDash [\![\phi(d) \equiv \phi(f)]\!]$. This would mean either $p'' \nvDash [\![\phi(d) \supset \phi(f)]\!]$ or $p'' \nvDash [\![\phi(f) \supset \phi(d)]\!]$. Without loss of generality, suppose it is the former case. Then, for some p''' accessible from p'', both $p''' \vDash [\![\phi(d)]\!]$ and $p''' \vDash [\![\sim\phi(f)]\!]$ hold. Now, because p''' is accessible rom p'', and $p'' \vDash [\![d \epsilon f]\!]$, $p''' \vDash [\![d \epsilon f]\!]$. Thus, by Theorem 2.7, and the assummption that $\phi(x)$ has only x free, $p''' \vDash [\![\phi(d)]\!] \equiv [\![\phi(f)]\!]$. Thus, because we have $p''' \vDash [\![\phi(d)]\!]$, it follows that $p''' \vDash [\![\phi(f)]\!]$. But this contradicts $p''' \vDash [\![\sim\phi(f)]\!]$.

Theorem 2.10

Here is why Theorem 2.10 follows from Lemma 2.9. Suppose $p \vDash (\forall x)[\![x \in f \equiv x \in g]\!]$ and $f, g \in R_\alpha^{\mathcal{G}}$. Then, by Lemma 2.9, $p \vDash [\![f \approx_\alpha g]\!]$, which implies $p \vDash [\![f \approx_{\alpha^+} g]\!]$, which implies $p \vDash f \approx_{\alpha^+} g]\!]$, which implies $p \vDash f \approx g$, which implies $p \vDash [\![f \approx g]\!]$.